

GUI approaches at ESS

Software for the science programme, from proposal to analysis

Andrew Sazonov

European Spallation Source (ESS)

Data Management and Scientific Computing (DMSC)

NOBUGS 2026 · GUI Workshop · 21 Sep 2026

One facility, two host countries

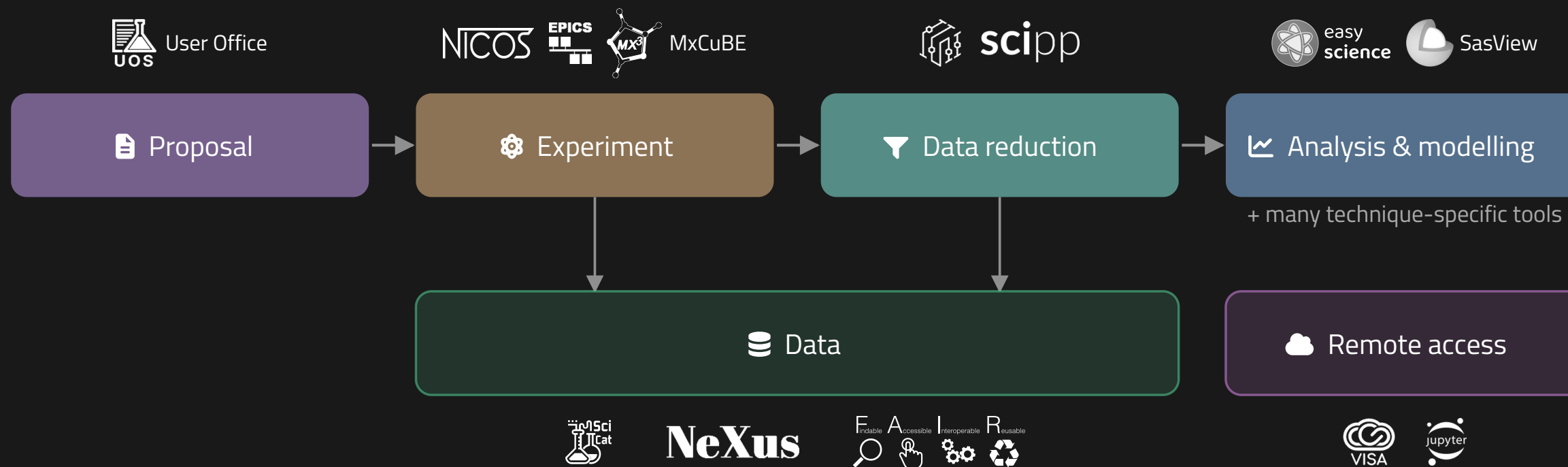
ESS is in Lund. DMSC is in Lyngby, near Copenhagen.



The integrated data pipeline

Different user interfaces support each stage of the data pipeline.

< 3 | 7 >



UI technologies in a nutshell

Qt for desktop applications. Web tools for other stages.

DESKTOP

Qt Widgets · PyQt6

 **NICOS**

nicos-controls.org

 MLZ

- Beamline control GUI
- ESS uses custom panels
- One layout per instrument

Qt Widgets · PySide6

 **SasView**

sasview.org

 many facilities

- Small-angle scattering analysis
- A new toolkit means a rewrite
- Adopted with the project

Qt QML · PySide6

 **EasyDiffraction · EasyReflectometry · EasyImaging**

easyscience.org

 U Göttingen  JCNS at MLZ  Chalmers

- Declarative UI in QML, separate from Python code
- EasyDynamics, EasyShapes and EasyTexture prototypes
- Prototypes become products without a rewrite

WEB

ipywidgets · Voilà

 **essreduce**

scipp.github.io/ess/reduce

 STFC

- Generated from a scipp workflow
- Documented, not deployed at ESS
- Works inside notebooks

Streamlit · Python

SANS-webapp

github.com/ai4se1dk/SANS-webapp

 AI4SE1DK

- SANS fitting
- AI assistant built in
- Little GUI code to maintain

Angular · TypeScript

 **SciCat**

scicatproject.org

 PSI  MAX IV

- Data catalogue
- Find and access datasets
- Browsed from any institution

React · TypeScript

 **User Office**

github.com/UserOfficeProject

 STFC  ELI  Diamond

- Proposal submission
- Facility access
- Used before anyone arrives

Challenges and controls

Most challenges are **organisational**. Beamline and accelerator controls use different UI approaches.

Organisational

- Little is shared between groups
 - Each group picks its own stack
- GUI expertise stays inside each group
 - No common review or style practice
- The same people do backend and GUI work

Technical

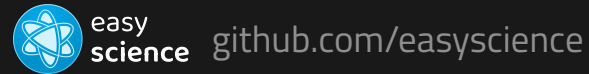
- Scientific plotting rarely supports QML
 - Embedding plots takes extra work
 - Embedded plots may not match the UI style
- Built-in QML charts lack some features
 - They would need work for scattering data
- ESS maintains a NICOS fork
 - We aim to converge with the main project

Beamline vs accelerator

- Beamline GUIs come from ECDC in DMSC
 - One shared NICOS client
 - Phoebus (Java desktop) for EPICS screens
- Accelerator GUIs come from elsewhere
 - Another division, another stack
 - No UI components in common

Reusable UI components

Shared QML components across EasyScience applications.



What we share

- The workflow bar, the panels and the controls
- Used by every EasyScience application
- **Shared only within EasyScience**
 - Other ESS projects use their own components

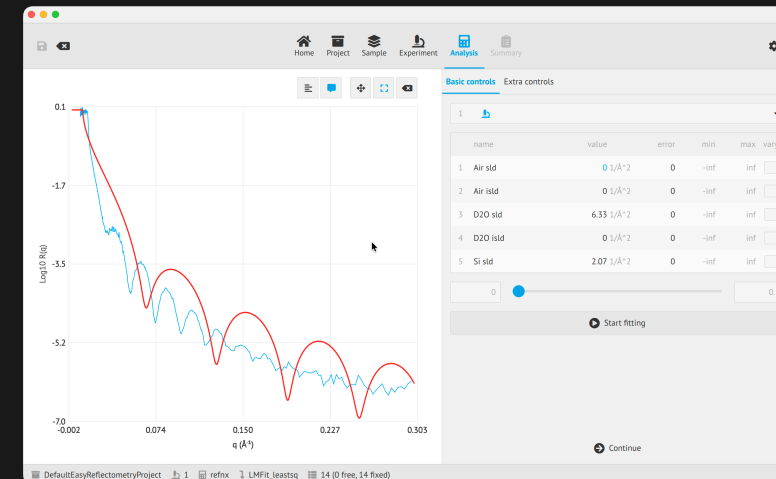
How we document it

- Five examples in the repository
- The applications are the main reference
 - The docs site is set up but not written
- **Next: write the docs and update them each release**

EasyDiffraction · diffraction data analysis



EasyReflectometry · reflectometry data analysis



UI generation and AI assistants

< 7 | 7

An AI assistant is available today. Two UI generation ideas are not built yet.

UI generation

 ideas

- Two directions we are considering
- We could generate data reduction notebooks for Voilà with AI
 - The workflow already describes the parameters
- We could assemble a new EasyScience GUI with AI
 - Using our shared components and existing applications as examples

AI assistants

SANS-webapp github.com/ai4se1dk/SANS-webapp

- Suggests a scattering model from the data
- Answers questions about the fit
- **Sets parameters and runs the fit when asked**
 - Its tools are off until you switch them on

SANS-webapp · AI-assisted model selection, with the chat panel on the right

